

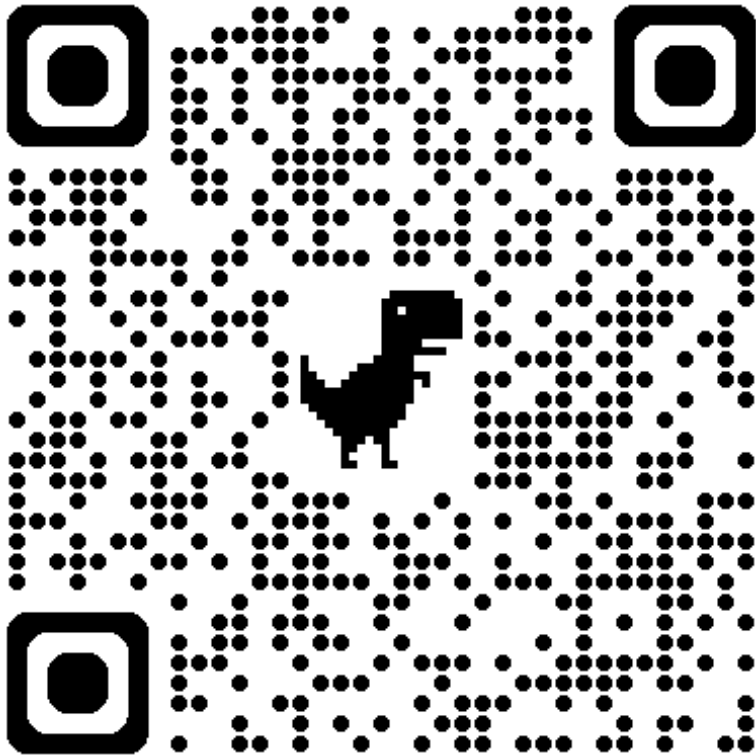
Introduction to Scientific Computing

ENGR1050 – Lecture 1

Fall 2025

Nat Trask

University of Pennsylvania



Computing

What is it



Course Organization

Syllabus, Schedule, Gameplan

Hello World

Submit a dummy assignment to
make sure you have a working
environment

Who am I?

About me



I joined Penn as an Associate Prof in August 2023



I was senior staff at Sandia National Laboratories 2016-2023, and hold a joint faculty appointment still



I earned my Ph.D. in Applied Mathematics at Brown University in 2015, and hold dual Bachelor degrees in Mechanical Engineering and Applied Mathematics from University of Massachusetts.



I've run a consulting simulation business since 2010.

My research focuses on integrating AI/ML with simulation to predict extreme physics

What is a national laboratory?

Department
of Energy
and the
National Lab
Complex

Office of Science Laboratories

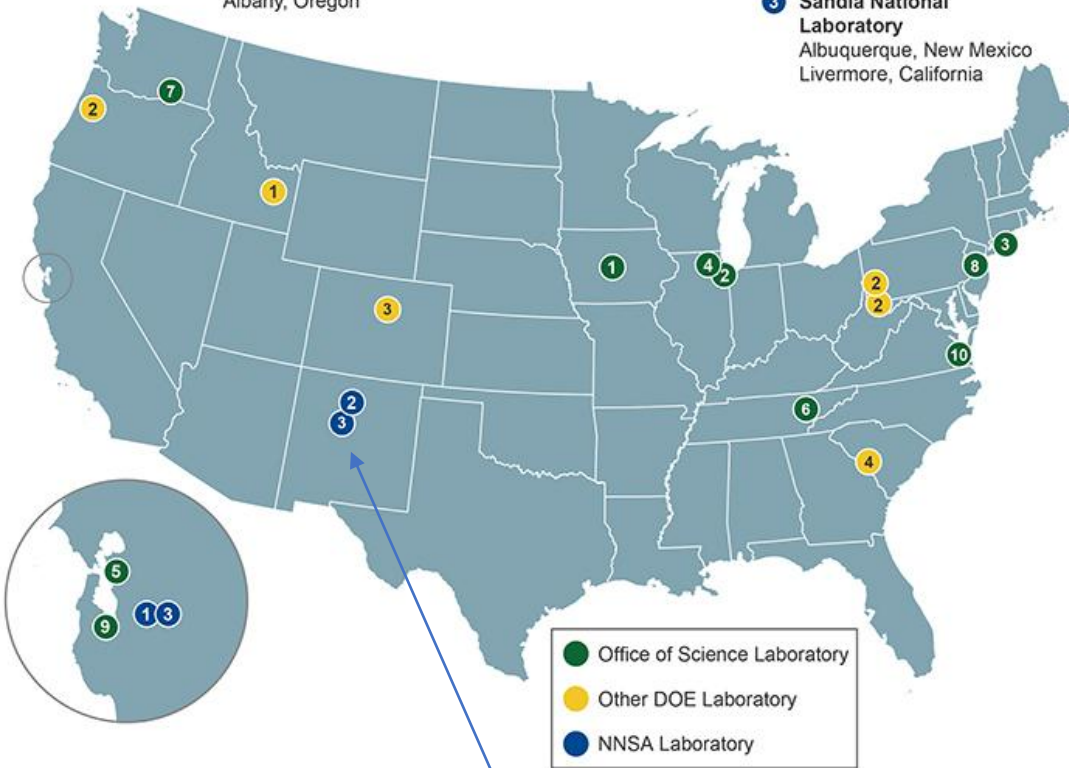
- 1 Ames Laboratory
Ames, Iowa
- 2 Argonne National Laboratory
Argonne, Illinois
- 3 Brookhaven National Laboratory
Upton, New York
- 4 Fermi National Accelerator Laboratory
Batavia, Illinois
- 5 Lawrence Berkeley National Laboratory
Berkeley, California
- 6 Oak Ridge National Laboratory
Oak Ridge, Tennessee
- 7 Pacific Northwest National Laboratory
Richland, Washington
- 8 Princeton Plasma Physics Laboratory
Princeton, New Jersey
- 9 SLAC National Accelerator Laboratory
Menlo Park, California
- 10 Thomas Jefferson National Accelerator Facility
Newport News, Virginia

Other DOE Laboratories

- 1 Idaho National Laboratory
Idaho Falls, Idaho
- 2 National Energy Technology Laboratory
Morgantown, West Virginia
Pittsburgh, Pennsylvania
Albany, Oregon
- 3 National Renewable Energy Laboratory
Golden, Colorado
- 4 Savannah River National Laboratory
Aiken, South Carolina

NNSA Laboratories

- 1 Lawrence Livermore National Laboratory
Livermore, California
- 2 Los Alamos National Laboratory
Los Alamos, New Mexico
- 3 Sandia National Laboratory
Albuquerque, New Mexico
Livermore, California

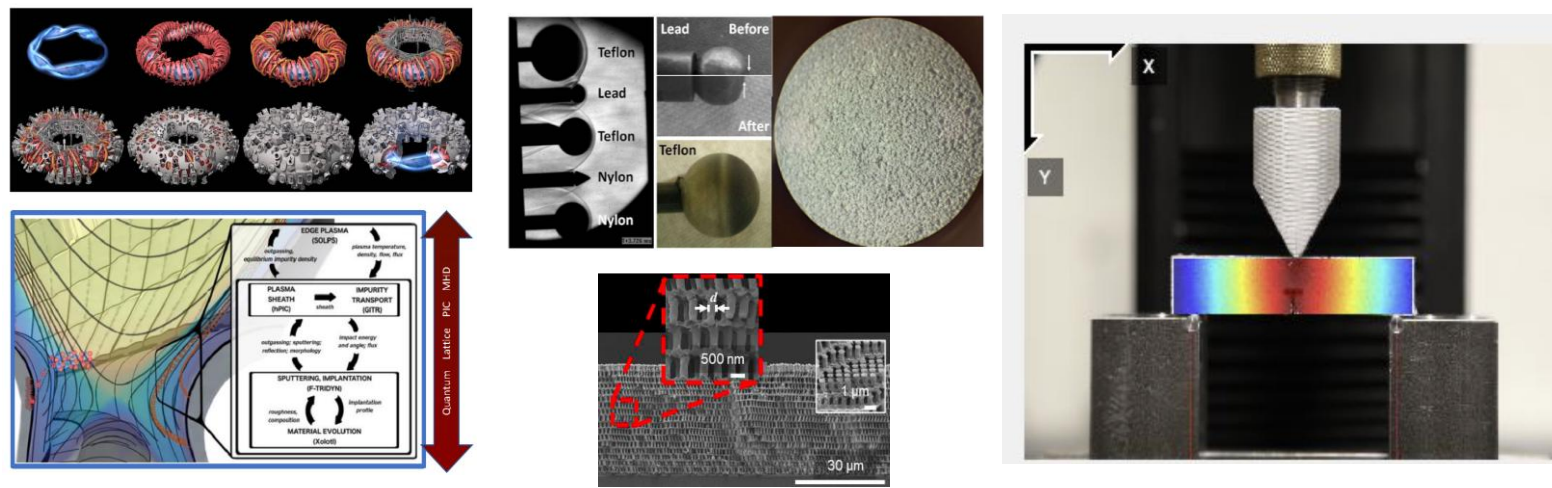


Sandia Albuquerque



Funders of science in US:
DoD \$120B/yr
NIH \$33B/yr
DoE \$19B/yr
NSF \$8B/yr
(2023 numbers)

My research: AI/ML for extreme physics systems



pimilab.github.io

source Scheduler... Grants TRASK, Nathaniel -... natrask/ENM5320 ENGR 1050-001 202... natrask/PIMI-Lab: L... Google AI studio OneStop Portal ed ENM 5320 001 – Ed...

 Physics-Informed Machine Intelligence Laboratory

Home Research Team Publications Join

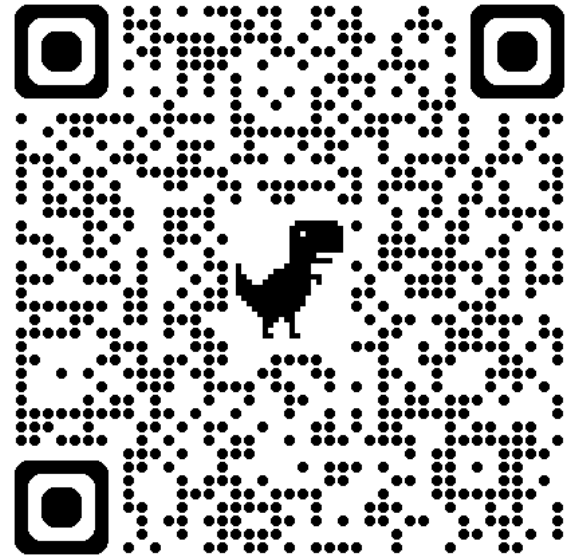
Welcome to the Physics-Informed Machine Intelligence Laboratory

We are a research group at [University of Pennsylvania](#) led by [Prof. Nat Trask](#).

We develop machine learning methods grounded in the **structure of physics and geometry**. By combining tools from geometric mechanics, exterior calculus, and variational modeling with modern AI architectures, we create **interpretable** and **reliable** models for complex physical systems in high-consequence engineering settings. Our work spans simulation, scientific discovery, and data-driven inference across multiscale

Who are you?

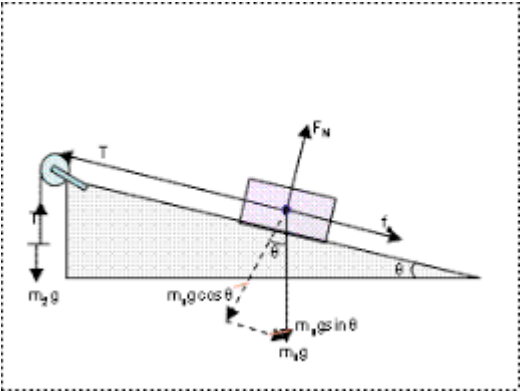
<https://forms.gle/Umy16mfbfLs4cYU9A>



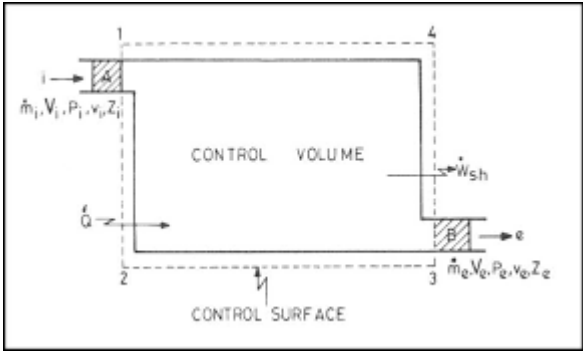
What is scientific computing, and why are we learning how to do it?

(It is not the same as normal computing)

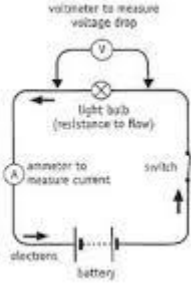
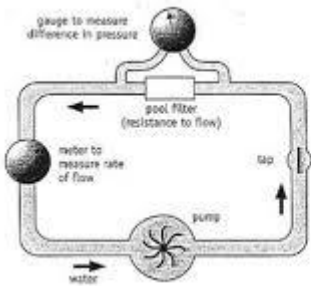
Scientific computing is programming *to solve models of physical systems*



Free body diagrams



Control volume analysis



Circuit models

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$
$$\rho \left[\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v + \frac{\partial u}{\partial z} w \right] = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho g_x$$
$$\rho \left[\frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} u + \frac{\partial v}{\partial y} v + \frac{\partial v}{\partial z} w \right] = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \rho g_y$$
$$\rho \left[\frac{\partial w}{\partial t} + \frac{\partial w}{\partial x} u + \frac{\partial w}{\partial y} v + \frac{\partial w}{\partial z} w \right] = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + \rho g_z$$

Differential equation models

During your degree, you will learn to build mathematical models that describe the behavior of physical systems and let you optimize their design

Analytical models vs experiments, computation, and statistical models

Analytical models are

Good at:

Giving closed form expressions
Interpretability
Exploring qualitative regimes

Bad at:

Handling real world complexity
**Limited by the math we can
work out by hand!**

Drag Equation

$$F_D = \frac{1}{2} \rho u^2 C_D A$$

The drag equation estimates the **resistance force** experienced by an object moving through a **fluid** due to **friction** and **pressure**.

F_D	drag force Force acting opposite to the direction of motion.
ρ	fluid density Mass per unit volume of the fluid.
u	flow velocity Speed of the fluid relative to the object.
C_D	drag coefficient Dimensionless number depending on shape and flow conditions.
A	area Reference area, usually the frontal area facing the flow.

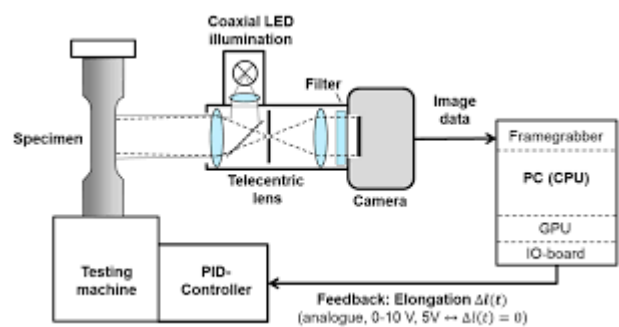
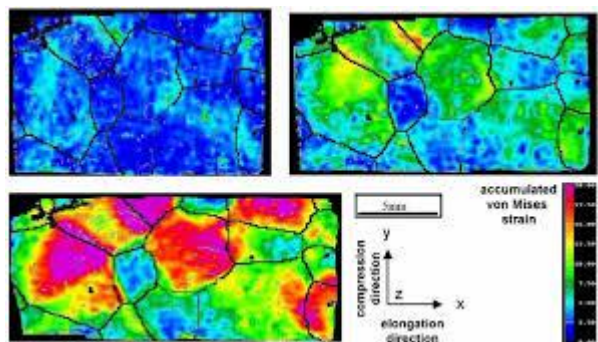
THEORY
Drag force is proportional to the fluid **density**, the square of the **velocity**, the **drag coefficient**, and the reference area.
It plays a key role in aerodynamics, hydrodynamics, and many engineering applications such as vehicle design, aircraft performance, and fluid machinery.


Ludwig Prandtl
(1875–1953)
German Physicist and Fluid Dynamicist

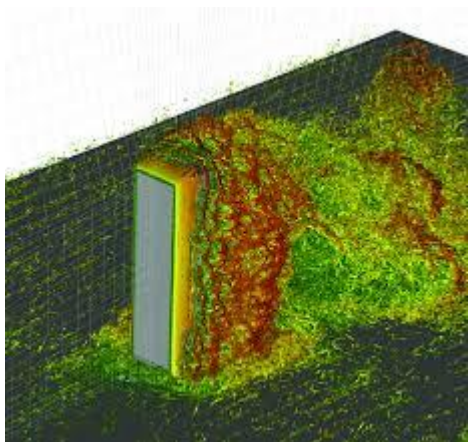
"Without friction, nothing would move."

Analytical models vs experiments, computation, and statistical models

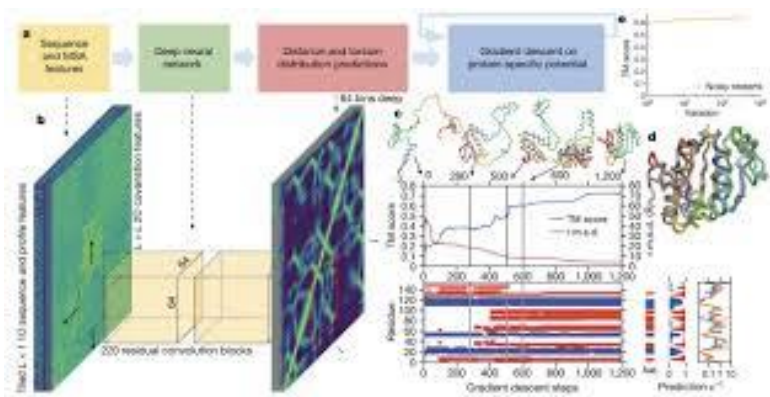
When we can't get an analytical model?



Experimental methods

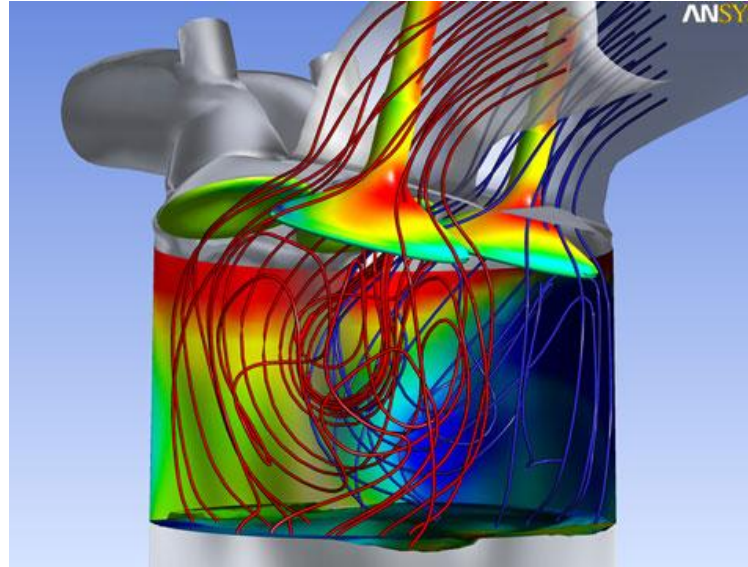


Computational methods



Statistical/AI methods

Scientific computing, from the big to the small



Microcontrollers

Small computers controlling engineering systems and robotics

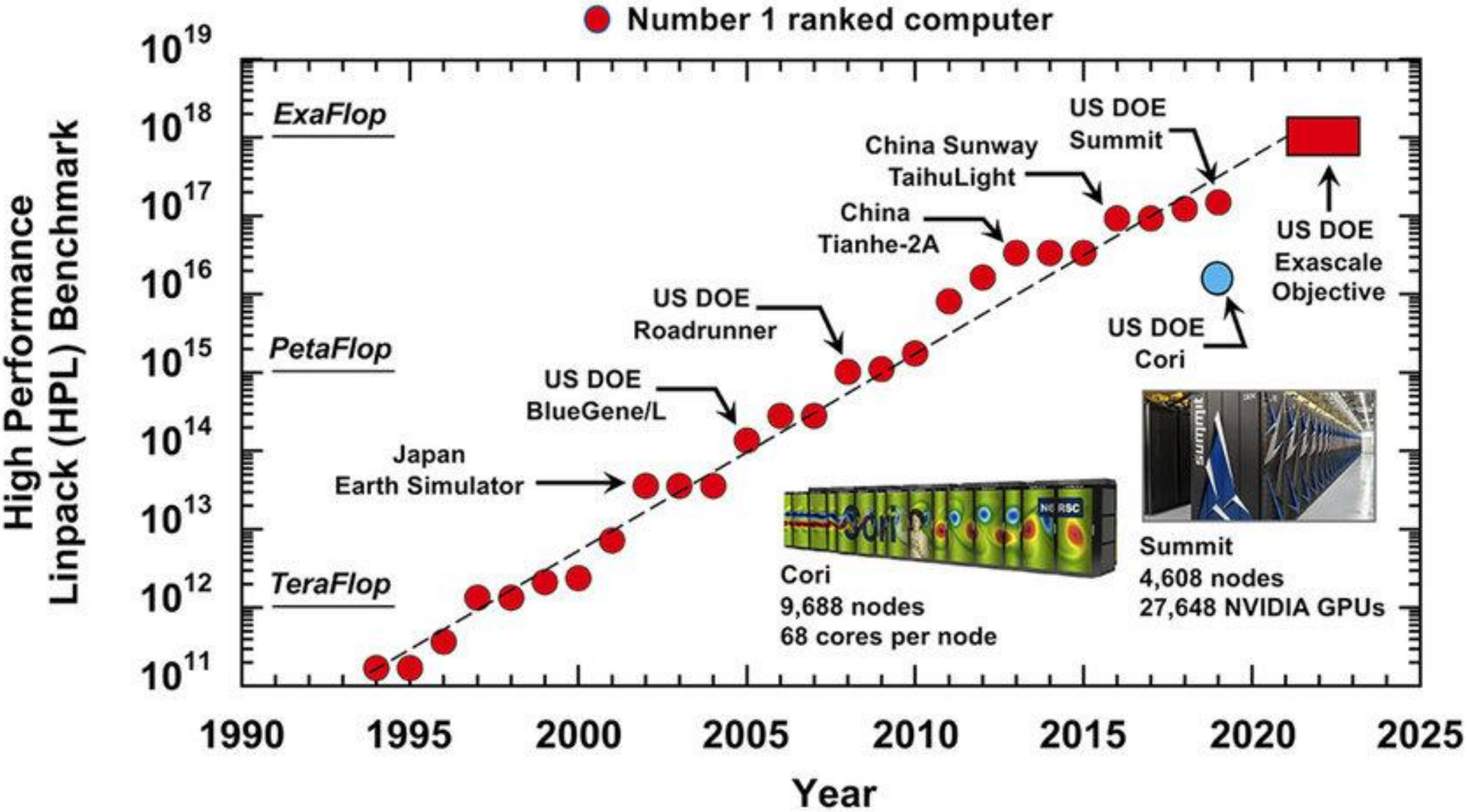
Industrial Computation

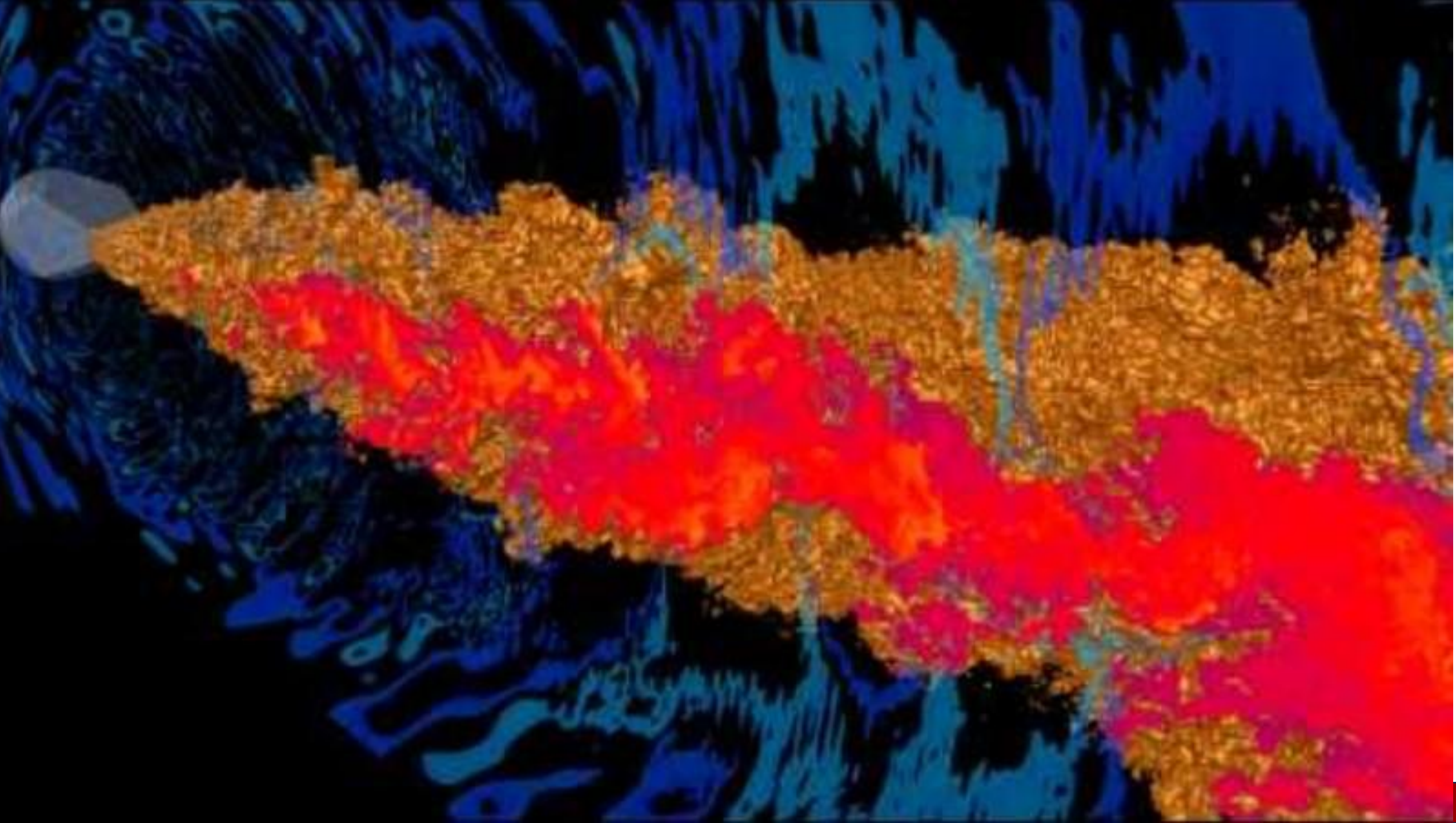
Simulations for industrial design that run on a desktop workstation

Supercomputing

Solving the biggest open problems in science + AI consuming the power of **16000 homes**

Big: Supercomputers represent the largest scientific problems that can be solved





Supercomputers represent the largest scientific problems that can be solved



Supercomputers represent the largest scientific problems that can be solved



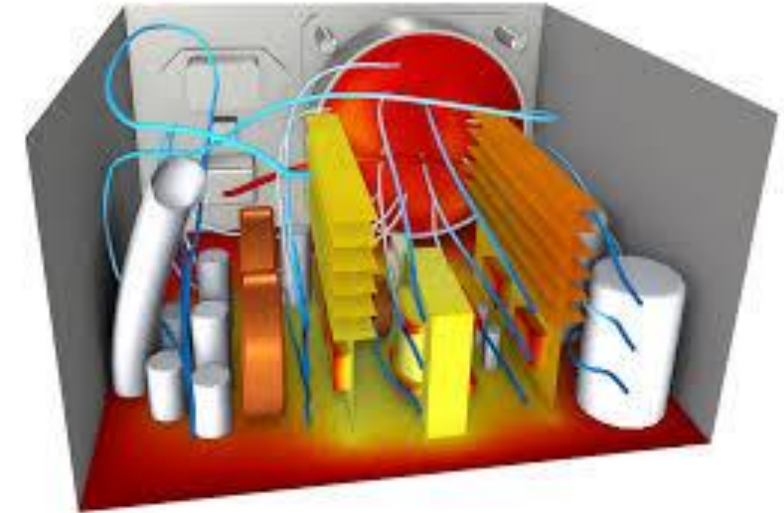
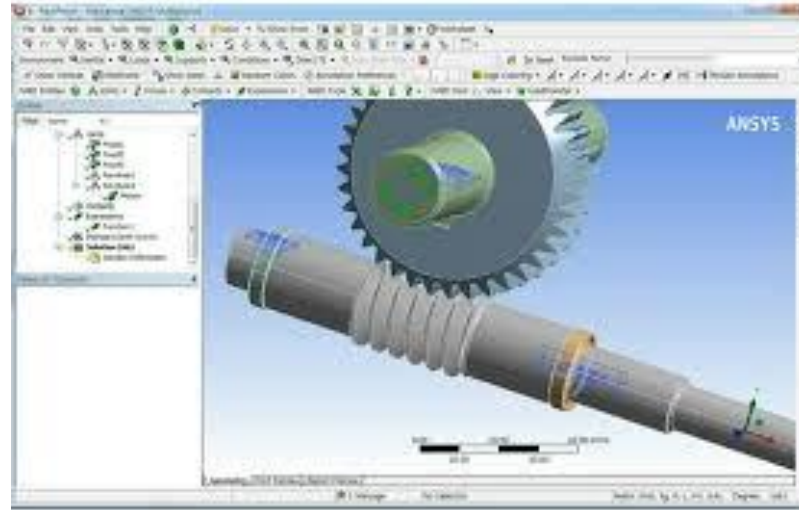
[Virtual tour of earthquake isolation system](#)



[Virtual tour of cooling and power plant](#)

Medium: computational fluid/solid mechanics + electromagnetism

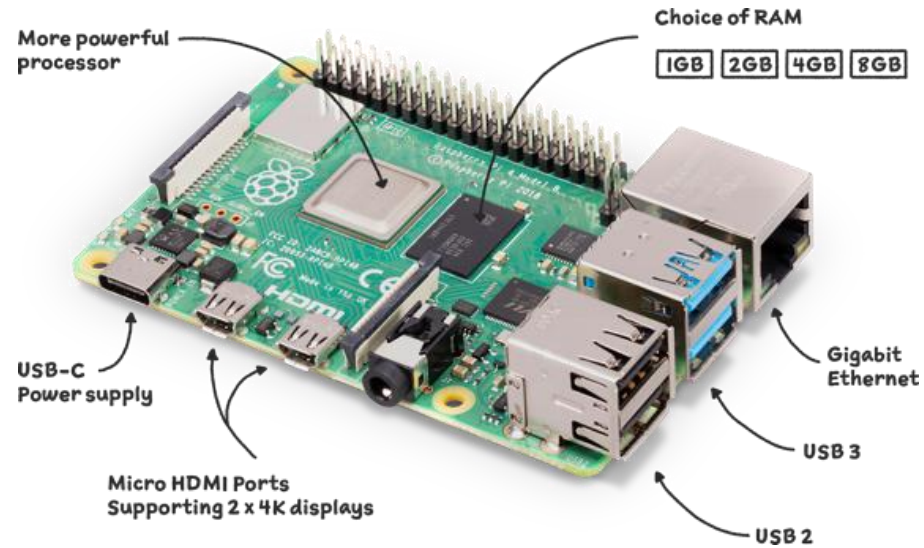
$$\underbrace{\frac{\partial \mathbf{u}}{\partial t}}_{\text{Variation}} + \underbrace{(\mathbf{u} \cdot \nabla) \mathbf{u}}_{\text{Convective acceleration}} = \underbrace{-\nabla w}_{\text{Internal source}} + \underbrace{\nu \nabla^2 \mathbf{u}}_{\text{Diffusion}} + \underbrace{\mathbf{g}}_{\text{External source}}.$$



Mathematical model describes flow
Conservation of
mass/momentum/energy

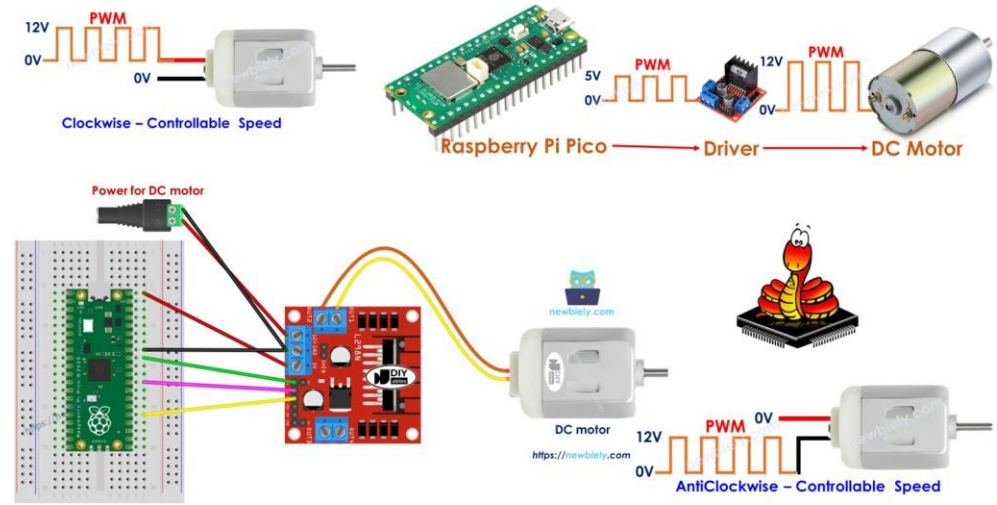
CAD geometry gives discrete model
Map equations onto something we can
solve on a computer

Small: Microcontrollers for robotics



Lightweight computers integrated with sensors

Useful for robotics, prototyping designs,



The future: AI for science and engineering



Massive AI models consuming a city's worth of power

AI-driven coding, scientific discovery, and beyond

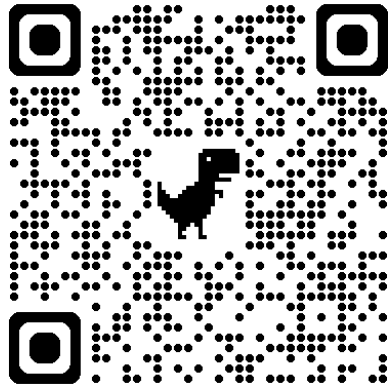


Compact AI chips small enough to fly on a drone

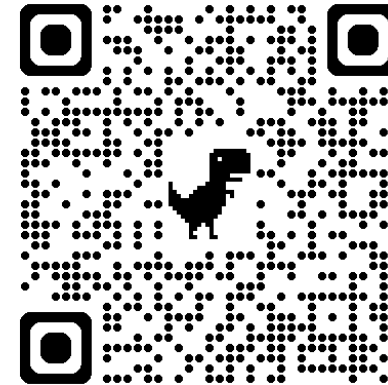
Smart sensing and control telling engineered systems what to do

How is this course organized?

All material is on the course Github:
<https://natrask.github.io/engr1050-fall2026>



All assignments are submitted through canvas:
<https://canvas.upenn.edu/courses/1950130/assignments/syllabus>



How is this course organized?

Part 1: Python basics

Goal: Comfortable reading and interpreting code

Exam 1

Part 2: Simulators and robots

Goal: Get practice deploying basics

Exam 2

Part 3: Machine learning + AI

Goal: Learn how AI works and how to use it to code

Exam 3

Final Project

Audio visualizer



6049
Digital microphone
~\$5

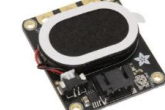


1528-4801-ND
Targetable LED strip
~\$13

Theramin



HC-SR04
Ultrasonic rangefinder
~\$7



3885
Speaker
~\$8

Weather station



BME280
Pressure/temp/
humidity sensor
~\$16



WEA012864DBPP3N00003
LCD display
~\$8



BME280
Pressure/temp/
humidity sensor
~\$16



**WEA012864DBP
P3N00003**
LCD display
~\$8



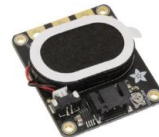
6049
Digital microphone
~\$5



HC-SR04
Ultrasonic rangefinder
~\$7



1528-4801-ND
Targetable LED strip
~\$13



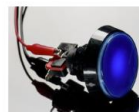
3885
Speaker
~\$8



SEN-101
Color detector
~\$8



SG90
Microservo motor
~\$4



Arcade button
Same as our buttons
~\$6

Types of lectures that we'll do

Coding demo days

- Slides introducing new concepts
 - Blackboard for derivations
- Live coding where I will step through how to approach a problem

Labwork days

- Meet in the GM lab in Towne
- Will be listed ahead of time on course calendar with reminder on Ed
 - Working with microcontrollers
- Some free lectures at end of semester to work on final projects w/ groups

Groupwork days

- Short guidance (<30m) on new material, what to do
 - Break into groups of 2-5
- I+TA will float around answering questions
 - Periodic check-ins to stay on task
- Submit work on canvas when complete

“Exam” days

- Closed book
- Problems taken primarily from HW and in-class exercises
 - Pencil and paper written exams
- Easy if you have done material on your own

Should I take this or should I take the CIS equivalent

Satisfies the same requirement for some majors

This class is more hands on
Tailored to specific needs of engineering majors

Scientific computing, differential equations, Arduino, hands-on

CIS1100/1200 sets up for higher-level CIS sequence

If you plan to take upper level CIS classes, ENGR1050 may not be best option

Happy to discuss if you have questions.
If you need a course substitution form filled, bring it to my office hours

CIS 1100 Introduction to Computer Programming

Introduction to Computer Programming is the first course in our series introducing students to computer science. In this class you will learn the fundamentals of computer programming, with emphasis on applications in science and engineering. You will also learn about the broader field of computer science and algorithmic thinking, the fundamental approach that computer scientists take to solving problems.

Fall or Spring

1 Course Unit

CIS 1200 Programming Languages and Techniques I

A fast-paced introduction to the fundamental concepts of programming and software design. This course assumes some previous programming experience, at the level of a high school computer science class or CIS 1100. (If you got at least 4 in the AP Computer Science A or AB exam, you will do great.) No specific programming language background is assumed: basic experience with any language (for instance Java, C, C++, VB, Python, Perl, or Scheme) is fine. If you have never programmed before, you should take CIS 1100 first.

Fall or Spring

1 Course Unit

Grading policy

30%

Homework Assignments

Five homework assignments combining short analytical problems with coding. The lowest homework grade is dropped.

30%

In-class Exercises

Short exercises completed during class and submitted on Canvas. The lowest two in-class exercise grades are dropped.

40%

Exams and Final Project

Three pencil-and-paper exams (one per unit) plus a mandatory final project. No exam or final project grade is dropped.

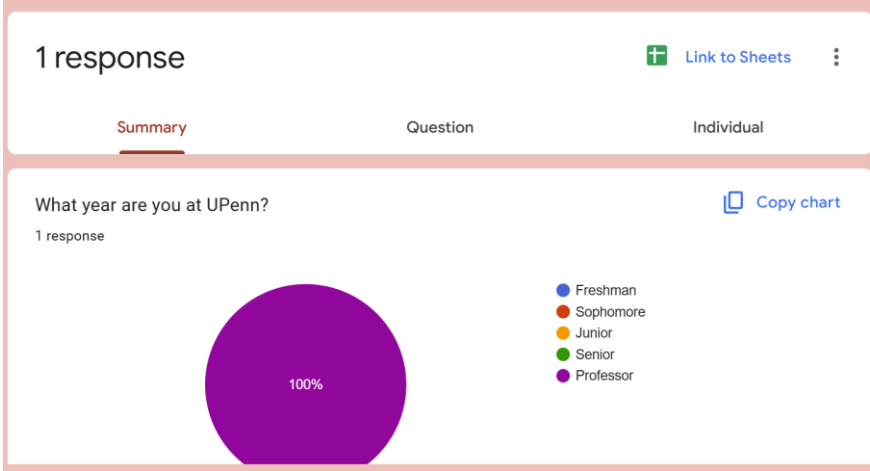
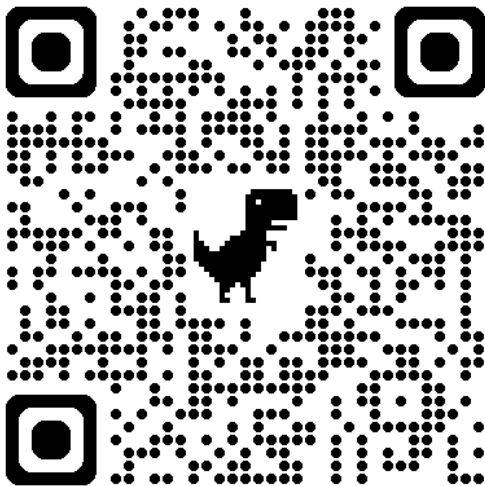
Communication with Prof. Trask outside of class

1. Office hours!

2. Surveys

3. Ed messages

(Do not count on me responding to any email)



Responses through surveys

A screenshot of an Ed Discussion page for the course ENM 5320 001. The page has a purple header with the course name and a search bar. On the left is a sidebar with a "New Thread" button and a list of categories: COURSES, DRAFTS, SCHEDULED, CATEGORIES, and SOCIAL. The main content area shows a list of threads. The thread "Assignment this week" is highlighted. To the right of the thread list is a detailed view of the "Assignment this week #53" thread. It shows the thread title, a heart icon, a comment icon, and a "73 VIEWS" count. The thread content is a message from "Anonymous" asking if there should be an assignment for the week. Below the message is a "1 Answer" section with a response from "Nat Trask" (marked as a staff member) explaining the situation and providing instructions. At the bottom of the answer section is a "Write another answer" button.

ed Penn ENM 5320 001 - Ed Discussion

New Thread

Search

Filter

General Tricia Lobo Smith 1

PINN-themed final project - Teammate Sea... General Hardik Shukla Smith

Neuroscience-themed final project - team... Assignments Cybele Smith Smith

Under the weather today - be at OH and at ... Lectures Cybele Smith Smith

Mar 30, 2025

Final Project Help Assignments Roy Wang Smith

"homework" posted General Nat Trask Staff Smith

Assignment this week Assignments Anonymous Smith 1

lecture tomorrow General Nat Trask Staff Smith 2

Absence from Class in May General Anindro Bhattacharya Smith 2

Class today General Nat Trask Staff Smith 4 2

Assignment this week #53

Anonymous 5 months ago in Assignments 73 VIEWS

Hi! Do we have an assignment for this week or should we focus on final projects? I didn't see one posted, so just wanted to check.

Comment Edit Delete Endorse ...

1 Answer

Nat Trask Staff 5 months ago

Yes let's pause on homework given the speedbump with the need for the repeated lecture. I'd like folks to identify their team for final projects - I'll add a placeholder assignment on canvas asking you to specify who you're working with and a one sentence blurb about what you want to work on - please complete by monday

Comment Edit Delete Endorse ...

Add comment

You have already written an answer to this question. Write another?

Write another answer

AI Policy

- Use of AI tools (e.g., GitHub Copilot, ChatGPT) is encouraged for learning and exploration. Work submitted should be your own.
- Be extremely careful of falling for the common pitfall of overestimating your command of the material if you aren't regularly working *without* AI.
- Exams are designed to make sure you know material without help from a LLM.
- All submitted work must be your own. If you use AI-generated code or text, you must clearly cite the tool and describe how you used it.
- Direct copying of solutions from AI or the internet without understanding is considered academic dishonesty. It is easy for us to tell when you have copy and pasted something from ChatGPT. If we ask you to explain why you wrote code a certain way and you cannot, that is a problem.
- If in doubt, ask the instructor for guidance on appropriate use of AI tools. We will devote a lecture or two to how to responsibly use AI.

Advice: Be patient, build muscle memory, and wait for the structured AI boot camp

Section 1: Basics

- Learn the basic Python syntax
- Do many exercises/etudes/drills to internalize patterns
 - **Beware this exam!**

Section 2:

- Build simple simulators
 - Build simple robots
- Get practice applying the basics
- **Hold off on AI** until you at least get through one round of programming!

Section 3:

- Open the AI floodgates
- **Implement the basic methods of AI**
- Learn how to responsibly use AI-assisted coding tools
 - Learn about pitfalls
- **Learn how to learn with AI**

```
tmux attach -t claudish
(0) + 6: 1 windows
(1) + aniflow: 4 windows
(2) + circl: 8 windows (attached)
(3) + claudish: 3 windows (attached)
(4) + main: 2 windows
(5) + passflow: 9 windows
(6) + lobehub-6: 1 windows (group lobehub: lobehub-6) (attached)
(7) + magai-7: 1 windows (group magai: magai-7) (attached)

claudish (sort: index)
  Bundled 488 modules in 53ms
  index.js 2.51 MB (entry point)
  • Build passes. Let me verify the key changes are
  • Read 1 file (ctrl+o to expand)
  • All changes look correct. Here's a summary of w
  ★ Insight
  Env-var gated provider visibility: The ALL_PROV
  - providers with an envVar requirement only app
  Custom) always show.
  Kimi Coding integr 1:main ee data structures
  - PROVIDER_MODEL_P s "kimi-coding" → "
  - getKnownModels(): fallback entry kc@kimi-for-
  - PROVIDER_FILTER_ALIASES: enables @kc shortcut
  Changes made to packages/cli/src/model-selector
  1. Env-var filtering - Renamed PROVIDER_CHOICES
  whose required env var isn't set. Affected pro
  - Gemini (GEMINI_API_KEY), OpenAI (OPENAI_API
  (KIMI_CODING_API_KEY), GLM (ZHIPU_API_KEY), GLM
  2. Kimi Coding - Added "kimi-coding": "kc@" pre
  3. GLM Coding Plan - Also added as a separate p
  • Now rebuild to make sure it's all clean, then s
  • Bash(bun run build 2>&1)
  | $ bun run build:cli && bun run build:bridge
  | $ cd packages/cli && bun run build
  | $ bun build src/index.ts --outdir dist --tar
  | ... +9 lines (ctrl+o to expand)
  • Build clean. Now let me stage all modified file
  • Bash(git status -u)
  | On branch main
  | Your branch is up to date with 'origin/main'
  | ... +62 lines (ctrl+o to expand)
  • Big changeset - 2:running ons from old src/,
  commit.
  • Bash(git add -A && git status --short | head -2
  | M .gitignore
  | M CLAUDE.md
  | M package.json
  | ... +17 lines (ctrl+o to expand)
  • Bash(git status --short | wc -l)
  | 179
  • 179 files staged. Now commit and tag.
  • Bash(git commit -m "feat!: v5.0.0 - composable
  -x@ 47 jack staff 1504 16 Feb 13:33 .
  nes (ctrl+o to expand)
  _projects (MCP)
  ization": {
  e": "Jack",
  nes (ctrl+o to expand)
  r 2 patterns (ctrl+o to expand)
  r: 400 {"error":{"error":{"message":"litellm.
  g_content is missing in assistant tool call message
  oup Fallbacks=None","type":"invalid_request_err
  LiteLLM kimi-for-coding · ~$0.038 · 99% (133/200k)
  ermissions on (shift+tab to cycle)
  session with:
  me f05ded15-9a20-4715-9ff7-c80fce90584b
  utting down proxy server...
  ne
  h main* 5m 13s }
  claudish 1: main 2: running 3: testing Copy 14:12 19-Feb
```

Rest of class, we will be working through a jupyter notebook



[←](#) [→](#) [↻](#) [🔍](#) [natrask.github.io/engr1050-fall2026/#schedule](#)

[Overview](#) [Schedule](#) [Homeworks](#) [Extras](#) [Grading](#) [Staff](#) [Pe](#)

The course meets twice per week. Most meetings combine a short lecture with live coding in a Jupyter notebook. Several meetings are dedicated lab sessions in the GM lab in the basement of Towne, where we will work hands-on with Raspberry Pi Picos and learn how to use them to drive sensors, motors, and simple mechatronic devices.

Course Schedule

Aug 26

Lecture 1: Introduction to Scientific Computing

[Notebook](#) [Open in Colab](#)

Group work submission on canvas

BAN_ENGR-1050-001 202530 > People > Groups

202530 (Fall 2025)

Home
Assignments
Discussions
Grades
People
Files
Syllabus
BigBlueButton
Collaborations
Search
Ed Discussion

Everyone Groups

Search Groups or People

+ Group

Lecture 01 1 Lecture 01	0 students	🔒
Lecture 01 2 Lecture 01	0 students	🔒
Lecture 01 3 Lecture 01	0 students	🔒
Lecture 01 4 Lecture 01	0 students	🔒
Lecture 01 5 Lecture 01	0 students	🔒

202530 (Fall 2025)

Home
Assignments
Discussions
Grades
People
Files
Syllabus
BigBlueButton
Collaborations

In class 01

Due Friday by 11:59pm Points 100 Submitting a file upload

Submit the completed jupyter notebook as a group.

File Upload

Upload a file, or choose a file you've already uploaded.

! Keep in mind, this submission will count for everyone in your Lecture 01 group.

Step 1: Join group

- Open canvas
- Click people tab
- Choose group (Lecture 1)
- Identify an empty group slot
- Have everyone in your group join the same one
 - You are now a group!

Step 2: Submit to canvas

- One group member needs to submit
 - Automatically distributed across whole group
- Double check that it shows up as submitted for everyone in group